

Agentic SDLC User Guide

Governed software delivery with Microsoft Foundry and Microsoft Agent Framework

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12-agent SDLC workflow				12 agents
Named Microsoft Foundry agents run automatically through APIM. The selected execution policy controls where the workflow pauses.				
 1. Requirements Agent Agent action · plan · 010-requirements-agent	 Last run 8/25/26, 7:27 PM	 Automated checkpoint Backlog Generation Approval · after Plan and Scope Approval	Backlog Generation Approval is already approved	
 2. Planning Agent Agent action · plan · 020-planning-agent	 Last run 8/25/26, 7:28 PM	 Automated checkpoint Backlog Generation Approval · after Plan and Scope Approval	Backlog Generation Approval is already approved	
 3. Architecture Advisor Agent Agent action · design · 030-architecture-advisor-agent	 Last run 8/25/26, 7:28 PM	 Automated checkpoint Architecture and Design Approval · after Backlog Generation Approval	Architecture and Design Approval is already approved	
 4. Code Generation Agent Agent action · build · 040-code-generation-agent	 Last run 8/25/26, 7:29 PM	 Automated checkpoint Code Generation Approval · after Architecture and Design Approval	Code Generation Approval is already approved	
 5. Test Planning Agent Agent action · test · 050-test-planning-agent	 Last run 8/25/26, 7:30 PM	 Automated checkpoint Test Acceptance Approval · after Code Generation Approval	Test Acceptance Approval is already approved	
 6. Testing Agent Agent action · test · 060-testing-agent	 Last run 8/25/26, 7:30 PM	 Automated checkpoint Test Acceptance Approval · after Code Generation Approval	Test Acceptance Approval is already approved	
 7. Test Automation Agent Agent action · test · 070-test-automation-agent	 Last run 8/25/26, 7:30 PM	 Automated checkpoint Test Acceptance Approval · after Code Generation Approval	Test Acceptance Approval is already approved	
 8. Security and Compliance Agent Agent action · security · 080-security-compliance-agent	 Last run 8/25/26, 8:09 PM	 Automated checkpoint Security Review Approval · after Test Acceptance Approval	Security Review Approval is already approved	
 9. DevOps / Release Agent Agent action · deploy · 090-devops-release-agent	 Last run 8/25/26, 8:09 PM	 Automated checkpoint Release and Deployment Approval · after Pull Request Review Approval + Security Review Approval	Release and Deployment Approval is already approved	
 10. Ops Monitoring Agent Agent action · operate · 100-ops-monitoring-agent	 Last run 8/25/26, 9:50 PM	 Automated checkpoint Operate and Improve Approval · after Release and Deployment Approval	Operate and Improve Approval is already approved	
 11. Knowledge Assistant Agent action · operate · 110-knowledge-assistant	 Last run 8/25/26, 9:51 PM	 Automated checkpoint Advisory — no publication gate · after Release and Deployment Approval	Runs automatically when eligible	
 12. Insights Agent Agent action · improve · 120-insights-agent	 Last run 8/25/26, 9:53 PM	 Automated checkpoint Operate and Improve Approval · after Release and Deployment Approval	Operate and Improve Approval is already approved	

The Agentic SDLC Software Factory uses 12 ordered Microsoft Foundry Prompt Agents (010 through 120), Microsoft Agent Framework orchestration, Azure API Management, and nine human approval gates. This guide follows one real project, **Retail Store Task Execution**, from intake to a published Angular/Ionic application and FastAPI service, then covers parallel project sets and the updated phone, iPad, and web workspaces.

Agent output is always a proposal first. Saving a draft does not write to Azure DevOps, GitHub, or Azure. External publication occurs only when an authorized human approves the matching gate.

Live Retail sample

Component	URL
Agentic SDLC project	https://agentic-sdlc-ui-my.azurewebsites.net/projects/f2cbe11d-1812-40e4-85c5-085f78daf72d
Azure DevOps project	https://dev.azure.com/csdmichael/Retail%20Store%20Task%20Execution
Azure DevOps backlog	https://dev.azure.com/csdmichael/Retail%20Store%20Task%20Execution/_backlogs/backlog
Azure DevOps test plan 426	https://dev.azure.com/csdmichael/Retail%20Store%20Task%20Execution/_testPlans/define?planId=426

Component	URL
Azure DevOps Project Overview wiki	https://dev.azure.com/csdmichael/05a6eec7-fc29-46c3-9b6f-2cdfa37fe6df/_wiki/wikis/01cb9f36-8e63-426e-b57d-91b623d3fc73?pagePath=%2FProject-Overview
GitHub repository	https://github.com/csdmichael/retail-store-task-execution
Pull request 1	https://github.com/csdmichael/retail-store-task-execution/pull/1
Successful feature-branch CI	https://github.com/csdmichael/retail-store-task-execution/actions/runs/32870761400
Successful Azure deployment	https://github.com/csdmichael/retail-store-task-execution/actions/runs/32873630512
Published UI	https://retail-store-task-execution-ui.azurewebsites.net
Published API health	https://retail-store-task-execution-api.azurewebsites.net/health
Swagger UI	https://retail-store-task-execution-api.azurewebsites.net/docs
OpenAPI document	https://retail-store-task-execution-api.azurewebsites.net/openapi.json

Sample inputs

The walkthrough uses all three source documents in `docs/samples/Retail Store Task Execution` :

Document	Purpose
<code>Retail Store Task Execution - Requirements.docx</code>	Business scope, roles, workflows, acceptance criteria, and non-functional requirements
<code>Retail Store Task Execution - Technical Requirements.docx</code>	Angular/Ionic UI, FastAPI services, data, identity, integration, and Azure constraints
<code>Retail Store Task Execution - UX Mockups.docx</code>	Screen inventory, interaction behavior, responsive expectations, and accessibility guidance

Each file is committed to the generated GitHub repository under `docs/intake/` . Text is extracted separately and supplied to agents as bounded, labelled context so all three sources contribute without exhausting the model budget.

Roles and accountability

Human role	Primary responsibility
Requester	Request access with identity and business justification
Business User or Product Owner	Submit scope and approve plan, backlog, and improvement outcomes
IT User, Architect, or Developer	Review architecture, generated code, test evidence, and release readiness

Human role	Primary responsibility
QA or Security Reviewer	Review automated evidence, controls, mitigations, and residual risk
App Owner	Manage users, permissions, connectors, agents, settings, audit, and authorized gate recovery
Guest	Read approved artifacts without write access

The FastAPI service enforces permissions and gate prerequisites. Disabled buttons are useful guidance, but they are not the security boundary.

The governed workflow

#	Agent	Proposal or evidence	Human checkpoint
010	Requirements Agent	Scope, actors, requirements, constraints	Backlog Generation Approval
020	Planning Agent	Epic, Features, User Stories, Tasks, estimates	Backlog Generation Approval
030	Architecture Advisor Agent	Architecture, dataflow, contracts, security design	Architecture and Design Approval
040	Code Generation Agent	Application scaffold, tests, CI/CD, pull request	Code Generation Approval
050	Test Planning Agent	Test strategy, plan, suites, and cases	Test Acceptance Approval
060	Testing Agent	Execution assessment and quality evidence	Test Acceptance Approval
070	Test Automation Agent	Automated-run evidence and test-plan publication	Test Acceptance Approval
080	Security and Compliance Agent	Prioritized controls, mitigations, residual risk	Security Review Approval
090	DevOps / Release Agent	Hosting, OIDC, merge, deployment, release record	Release and Deployment Approval
100	Ops Monitoring Agent	Health indicators, alerts, dashboards, triage	Operate and Improve Approval
110	Knowledge Assistant	Advisory operational and onboarding brief	Advisory after release
120	Insights Agent	Metrics, thresholds, risks, improvement backlog	Operate and Improve Approval

All model calls travel through the configured APIM gateway to the named Foundry agents. Microsoft Agent Framework owns workflow topology, prerequisite checks, approval pause/resume, and checkpoints.

Parallel project sets and device layouts

Use **New Project Set** when several independent applications should enter the SDLC together. The wizard accepts 2 to 20 projects. Each project requires a functional requirements document; UX mockups and technical requirements are optional. Every project defaults to **Autonomous**, while Minimal review and Human review can be selected independently.

On a phone, project entries are stacked and the fixed bottom workspace keeps Dashboard, Existing Projects, Approvals, Workflow Runs, and the full menu within thumb reach.

SDLC Factory

MY

New Project Set

Step 2 of 4 · Project intake

Project intake

• 2 projects

1 Project

Name *

Description

Business owner

owner@example.com

Workflow execution

Autonomous ▼

Functional requirements *

Choose File

No file chosen

UX mockups

Choose File

No file chosen

Technical requirements

Dashboard

Existing Proje...

Approvals

Workflow Runs

More

Agent participation and model deployments are selected once for the set. Use a project's **Override** button to clone and replace the shared configuration for that project only. Other projects continue to inherit the set policy.

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App Owner

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Agentic SDLC

Software Factory

AGENTIC SDLC OPERATIONS

Dashboard

New Project

New Project Set

Existing Projects

Workflow Runs

Human Approval Queue

Agent Activity

Audit Trail

ADMIN

User Management

Role Permissions

Agent Configuration

Global Settings

APIM & Config

DOCUMENTATION

Overview

Architecture

Human-in-the-loop

Agent Responsibilities

Security & Guardrails

Michael Yaacoub · Sr Solution Engineer

New Project Set

Step 3 of 4 · Agent configuration

Agent configuration

2 projects

Shared configuration

Applied to every project unless overridden.

0 model overrides

☒	Requirements Agent plan	model-router (balanced) ▾
☒	Planning Agent plan	model-router (balanced) ▾
☒	Architecture Advisor Agent design	gpt-5.4 ▾
☒	Code Generation Agent build	gpt-5.1-codex ▾
☒	Test Planning Agent test	model-router (balanced) ▾
☒	Testing Agent test	gpt-5.1-codex ▾
☒	Test Automation Agent test	gpt-5.1-codex ▾
☒	Security and Compliance Agent security	gpt-5.4 ▾
☒	DevOps / Release Agent deploy	gpt-5.4 ▾
☒	Ops Monitoring Agent operate	model-router (balanced) ▾
☒	Knowledge Assistant operate	model-router (balanced) ▾
☒	Insights Agent improve	model-router (balanced) ▾

Back

Project overrides

Keep shared settings or replace them for one project.

Untitled project

Shared configuration

Override

Untitled project

Shared configuration

Override

Continue

ZIP import is the faster alternative. Put one project in each top-level folder. Include one functional requirements file and optionally one UX mockup and one technical requirements file. An optional `project.json` can provide the project name, owners, environment, execution mode, and explicit document paths. The API validates the complete archive before provisioning starts.

The confirmation step shows every project, its execution mode, shared-agent policy, and the fact that all workflow runs launch independently in parallel.

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Software Factory

AGENTIC SDLC OPERATIONS

Dashboard

New Project

New Project Set

Existing Projects

Workflow Runs

Human Approval Queue

Agent Activity

Audit Trail

ADMIN

User Management

Role Permissions

Agent Configuration

Global Settings

New Project Set

Step 4 of 4 · Confirm & launch

Confirm & launch

4 projects

4
Projects

12
Shared agents

0
Shared model overrides

Parallel
Workflow launch

1
Construction Site Progress Capture
myaacoub@microsoft.com · Dev

Autonomous

Shared models

2
Equipment Calibration Compliance
myaacoub@microsoft.com · Dev

Autonomous

Shared models

3
Field service Work Orders
myaacoub@microsoft.com · Dev

Minimal review

Shared models

4
Fleet Inspection And Maintenance
myaacoub@microsoft.com · Dev

Human review

Shared models

Back

Create set & launch

After confirmation, the API creates a distinct workflow run and checkpoint set for every project and advances eligible runs concurrently. Existing Projects opens automatically, groups projects by workflow-aware status, and shows stage progress, environment, owner, project-set membership, and drill-through.

- Awaiting Approval

Stage progress updates while Foundry agents are active. Running cards use an animated progress sheen and status pulse; the caption identifies the current stage and how many agents in that stage have completed.

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Existing Projects

4 project(s) · 4 shown

 Search by name, owner

Status
All statuses

Project set
Enterprise Demo Projec...

Failed 3

Field service Work Orders

Enterprise Demo Project Set

No description

Build Dev Myaacoub@Microsoft.Com

Workflow progress 38%

Build · workflow failed

Equipment Calibration Compliance

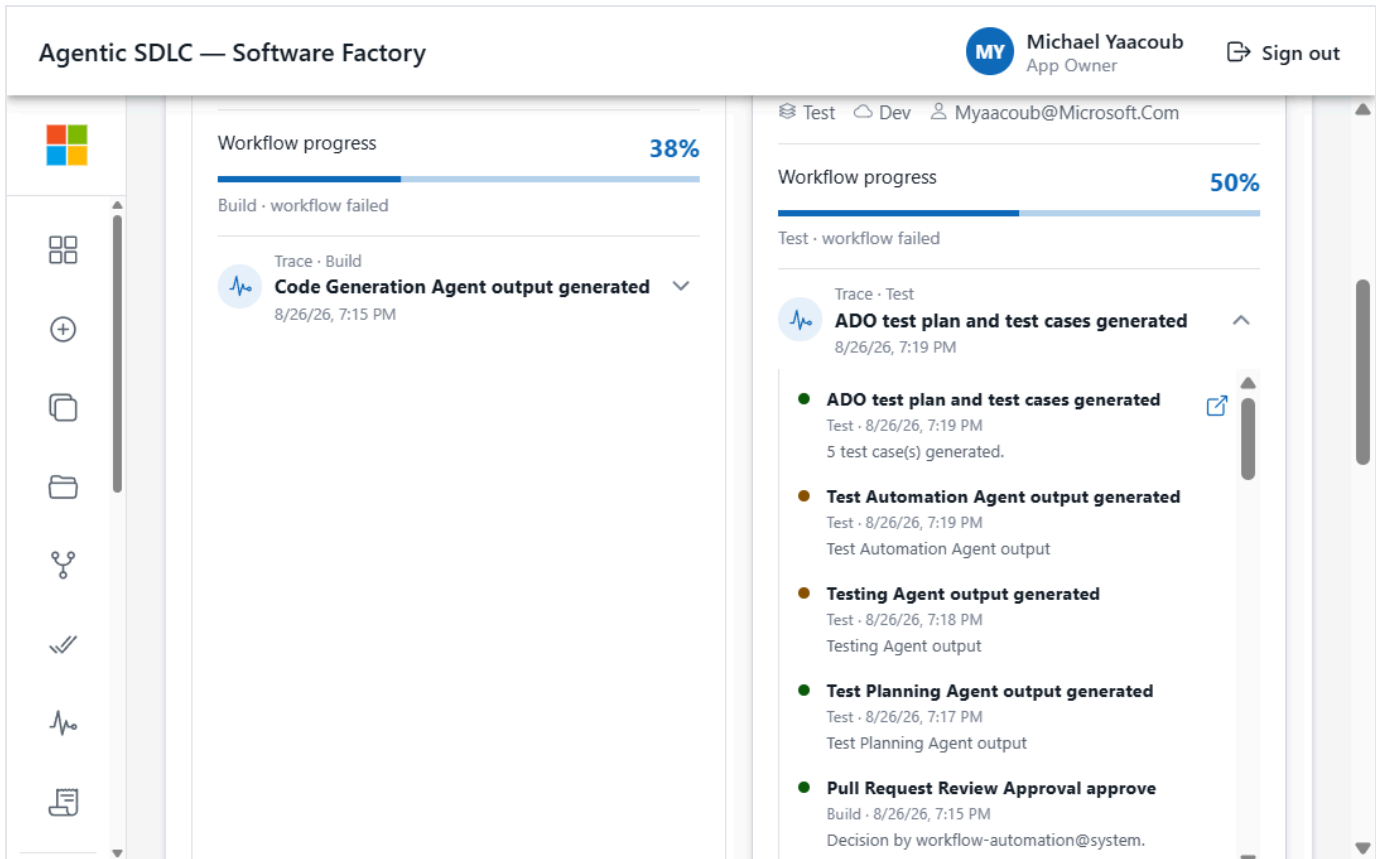
Enterprise Demo Project Set

No description

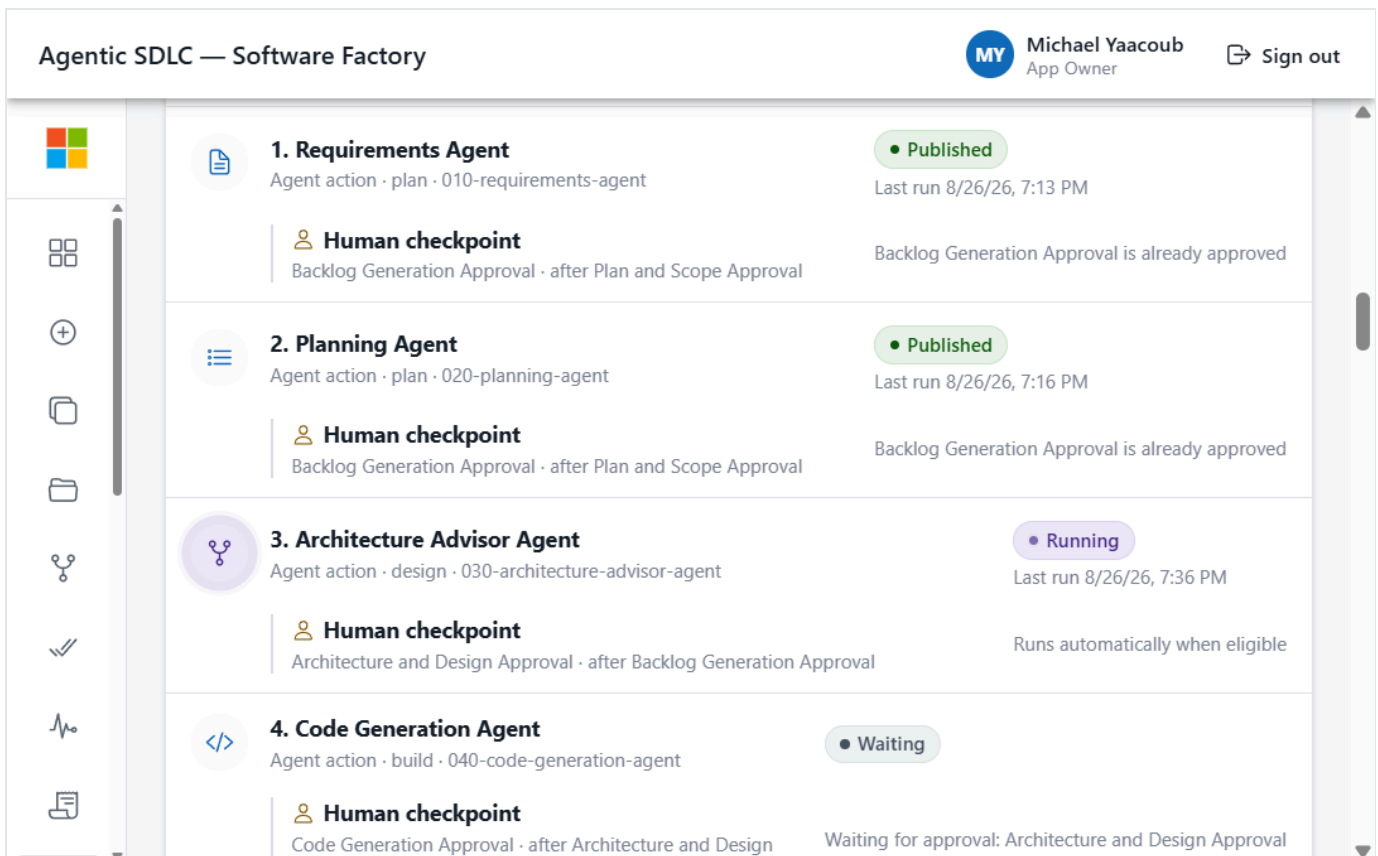
Test Dev Myaacoub@Microsoft.Com

Workflow progress 50%

Each card also exposes a **Trace** control. Its collapsed state shows the latest message and current stage. Expanding it reveals the chronological project trace, including ADO/GitHub provisioning, generated work items, code and pull-request evidence, state transitions, pipeline outcomes, Azure publication, wiki updates, and workflow completion, with direct links to external evidence.



Project Details uses the same live state. The active agent receives a restrained activity ring while queued and completed agents remain visually stable.



The iPad layout uses a persistent icon rail and two-column work area. Web uses the full grouped sidebar and information-dense panels. All three layouts expose the same server-authorized actions and records.

End-to-end walkthrough

1. Request access

Use Microsoft Entra ID, email OTP, or read-only guest access. A new requester can submit a work email and business reason for App Owner review.

Michael Yaacoub | Sr Solution Engineer — [LinkedIn](#) · [csdmichael](#)

Sign in

Enter your email address and we'll send you a one-time passcode. Accounts on **MngEnvMCAP829495.onmicrosoft.com** use Microsoft Entra ID single sign-on instead. Or continue as a guest for read-only access.

Email address

 you@company.com

Quick fill:

- @MngEnvMCAP829495.onmicrosoft.com
- @microsoft.com

Sign in

Enter your email address and we'll send you a one-time passcode. Accounts on **MngEnvMCAP829495.onmicrosoft.com** use Microsoft Entra ID single sign-on instead. Or continue as a guest for read-only access.

Email address

✉ you@company.com

Quick fill: • @MngEnvMCAP829495.onmicrosoft.com

2. Assign a least-privileged role

The App Owner creates or updates the user, selects the authentication provider, and assigns the role that matches the person's responsibilities.



User Management

App Owner only. Add or edit users and set each user's authentication method.

Add or edit user

Saving an existing email updates that user in place.

Name

Jane Doe

Email

name@company.com

Role

Business User ▼

Authentication method

Entra ID ▼

3. Define the Retail project

Enter the project outcome, accountable Business and IT owners, and target environment. The Retail sample focuses on prioritized store tasks, photographic proof, offline capture, exception handling, and same-day district compliance visibility.

New Project

Step 1 of 6 — Project details

Project details

Who owns this work and what is it called.

Project details

Project name *

Retail Store Task Execution

Description

Prioritised store task execution with photographic proof, offline capture, exception handling, and same-day district compliance visibility.

Business owner *

myaacoub@microsoft.com

IT owner

myaacoub@microsoft.com

4. Upload all three intake documents

Requirements are required. Technical requirements and UX mockups are optional but strongly recommended. Each upload is limited to 25 MB. Do not place passwords, tokens, keys, or connection strings in documents or prompts.

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Requirements & UX mockups

Where the agents should read intake material from.

Requirements & UX mockups

Word templates

Download a template, replace its placeholders, then upload the completed document below.

Requirements template

Functional and non-functional requirements

UX mockups template

Screen specifications and navigation flow

Technical requirements template

Architecture, stack, security, and delivery

Requirements document *

Choose File

Retail Store Task Execution - Requirements.docx

UX mockup document

Choose File

Retail Store Task Execution - UX Mockups.docx

Technical requirements document

Choose File

Retail Store Task Execution - Technical Requirements.docx

5. Confirm Azure DevOps and GitHub targets

The wizard derives a private ADO project and a GitHub repository from the project name. Visibility and target names remain visible before submission.

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Azure DevOps organization

https://dev.azure.com/csdmichael

Azure DevOps project

Retail Store Task Execution

Azure DevOps project visibility

Private

GitHub repository

csdmichael/retail-store-task-execution

GitHub repository visibility

Public ▼

①

Azure DevOps projects are always private because the organisation does not permit public projects. GitHub repositories can be public or private; in production the recommended default is private.

Target environment

Dev ▼

Back

Continue

6. Review systems of record

Confirm the effective work-item, test-management, source-control, build, and deployment systems. Project overrides remain explicit and auditable.

Systems of Record

Pre-populated from the global settings. Toggle **Override** on any row this project should track somewhere else; untouched rows keep inheriting the global value.

- ☐ Documentation
Inherited: GitHub docs/ folder · <https://github.com/csdmichael>
- ☐ Work Items Tracking
Inherited: Azure DevOps · <https://dev.azure.com/csdmichael>
- ☐ Test Cases & Test Plans Tracking
Inherited: Azure DevOps · <https://dev.azure.com/csdmichael>
- ☐ Code Build / Pipelines
Inherited: Azure DevOps · <https://dev.azure.com/csdmichael>
- ☐ Source Code Org
Inherited: GitHub · <https://github.com/csdmichael>

Back

Continue

7. Select the complete agent workflow

All 12 agents are selected for this run. Future agents remain disabled until every configured prerequisite gate is approved.

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test · gpt-4.1 · approval: admin

☒

DevOps / Release Agent

deploy · gpt-4.1 · approval: it_user

☒

Security and Compliance Agent

security · gpt-4.1 · approval: admin

☒

Ops Monitoring Agent

operate · gpt-4.1 · approval: it_user

☒

Knowledge Assistant

operate · gpt-4.1

☒

Insights Agent

improve · gpt-4.1 · approval: business_user

Back

Continue

8. Confirm and submit

The final wizard step summarizes ownership, environment, all three files, target systems, and selected agents. Submission provisions the project systems and creates the workflow run.

Technical requirements	Retail Store Task Execution - Technical Requirements.docx
UX mockup	Retail Store Task Execution - UX Mockups.docx
ADO	Retail Store Task Execution
ADO visibility	private
GitHub	csdmichael/retail-store-task-execution
GitHub visibility	public
Environment	Dev
Systems of Record overridden	None (all inherited)
Agents selected	12

ⓘ Submitting creates the project and opens the **Plan and Scope** approval gate. Agents will not act until it is approved.

Back

Create & submit for approval

9. Verify provisioned systems

The project record links the generated ADO project and GitHub repository. A failed connector must be repaired before an agent can publish to it.

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App Owner

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Systems of Record for this project

Where each class of SDLC output is tracked. Overridden rows are marked.

Documentation

• GitHub docs/ folder

https://github.com/csdmichael

>

Work Items Tracking

• Azure DevOps

https://dev.azure.com/csdmichael

Project: Retail Store Task Execution

>

Test Cases & Test Plans Tracking

• Azure DevOps

https://dev.azure.com/csdmichael

Project: Retail Store Task Execution

>

Code Build / Pipelines

• Azure DevOps

https://dev.azure.com/csdmichael

Project: Retail Store Task Execution

>

Source Code Org

• GitHub

https://github.com/csdmichael

>

10. Review the project overview

The project page displays owners, environment, current stage, systems of record, lifecycle, approval gates, and the 12-agent lane in one operating view.

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Retail Store Task Execution

Prioritised store task execution with photographic proof, offline capture, exception handling, and same-day district compliance visibility.

Awaiting Approval

Overview

Intake details captured when the project was created.

BUSINESS OWNER myaacoub@microsoft.com	IT OWNER myaacoub@microsoft.com	ENVIRONMENT Dev
CURRENT STAGE Plan	AZURE DEVOPS Retail Store Task Execution	ADO VISIBILITY private
GITHUB REPO retail-store-task-execution	GITHUB VISIBILITY public	

12-agent SDLC workflow

Named Microsoft Foundry agents generate proposals through APIM. Human checkpoints publish approved outputs.

12 agents

1. Requirements Agent

Agent action · plan · requirements-agent

Waiting

11. Approve Plan and Scope

The first gate confirms that the submitted business outcome and source material are suitable for agent work. Approval unlocks Requirements and Planning; it does not publish backlog items by itself.

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App Owner

Sign out

					Generate & Submit		
Architecture and Design Approval	Design	it_user	—	Generate proposal first	Awaiting Approval	Request changes	Reject
					Delegate		
					Generate & Submit		
Backlog Generation Approval	Plan	business_user	—	Generate proposal first Run the Planning Agent and select its reviewed hierarchy before approving backlog generation.	Awaiting Approval	Request changes	Reject
					Delegate		
					Generate & Submit		
Code Generation Approval	Build	it_user	—	Generate proposal first	Awaiting Approval	Request changes	Reject
					Delegate		
					Generate & Submit		
Pull Request Review Approval	Build	it_user	—	Not required Approve Code Generation first so a real Github pull request exists before PR review approval.	Awaiting Approval	Request changes	Reject
					Delegate		
					Generate & Submit		
Security Review Approval	Security	admin	—	Generate proposal first	Awaiting Approval	Request changes	Reject
					Delegate		
					Generate & Submit		
Test Acceptance Approval	Test	it_user	—	Generate proposal first	Awaiting Approval	Request changes	Reject
					Delegate		
					Generate & Submit		
Release and							

12. Review Requirements and Planning proposals

Long proposals begin in a compact three-row summary. Reviewers can show the full proposal, edit it, save a draft, and select exactly which artifacts are eligible for approval.

☒ Include in approval

Planning Agent
 Awaiting Approval
 Approve & publish

Proposal title
 Planning Agent output

Editable proposal
 Below is a **complete, reviewable Azure DevOps (ADO) hierarchy** for the "Retail Store Task Execution" project, structured for traceability, prioritization, acceptance criteria, and explicit linkage to intake documents. This hierarchy is ready for review and approval, and aligns with the requirements, technical constraints, and UX specifications

Summary view
 Show full proposal

Expand all
 Collapse all
 Select all
 Clear all
 Add work item
 Remove selected

> ☒ Epic Retail Store Task Execution delivery 3 children
 > ☒ Feature Epic 2 children
 > ☒ User Story **Outcome:** Head office tasks delivered to stores with clear priority and proof r... 3 children
 > ☒ Task Design: **Outcome:** Head office tasks delivered to stores with clear priority and proof re...
 > ☒ Task Implement: **Outcome:** Head office tasks delivered to stores with clear priority and proo...
 > ☒ Task Test: **Outcome:** Head office tasks delivered to stores with clear priority and proof requi...

13. Review the ADO hierarchy

Planning output appears as a collapsed Epic to Feature to User Story to Task tree. Titles, types, indentation, and child counts remain visible while detail editors stay closed. Selecting a child work item does not automatically select its parents. Clearing a parent clears its branch. Work-item deletion preserves the current review position and keyboard focus.

The approved Retail hierarchy created:

- 1 Epic
- 3 Features
- 6 User Stories
- 18 Tasks
- 3 dated sprints
- 6 shared queries
- 2 dashboards
- 1 delivery plan

>	✓	Epic	Retail Store Task Execution delivery	3 children
>	✓	Feature	Epic	2 children
>	✓	User Story	**Outcome:** Head office tasks delivered to stores with clear priority and proof r...	3 children
>	✓	Task	Design: **Outcome:** Head office tasks delivered to stores with clear priority and proof re...	
>	✓	Task	Implement: **Outcome:** Head office tasks delivered to stores with clear priority and proo...	
>	✓	Task	Test: **Outcome:** Head office tasks delivered to stores with clear priority and proof requi...	
>	✓	User Story	**Business Owner:** Director of Retail Operations	3 children
>	✓	Task	Design: **Business Owner:** Director of Retail Operations	
>	✓	Task	Implement: **Business Owner:** Director of Retail Operations	
>	✓	Task	Test: **Business Owner:** Director of Retail Operations	
>	✓	Feature	Features	2 children
>	✓	User Story	**Traceability:** [Requirements](https://github.com/csdmichael/retail-store-task-...	3 children

Backlog Generation Approval requires the Planning Agent artifact. A Requirements summary alone cannot satisfy the gate.

14. Review generated code through a pull request

Code Generation produced branch `feature/agent/e724c65d` and pull request 1 with the UI, API, database assets, tests, CI, and Azure deployment workflow. Pull Request Review cannot be approved until a real pull-request URL is present.

The screenshot shows a GitHub pull request interface. At the top, the repository name is 'csdmichael / retail-store-task-execution' with a 'Public' label. Navigation tabs include 'Code', 'Issues', 'Pull requests' (which is active and shows '1'), 'Actions', 'Projects', and 'Security and quality'. The pull request title is 'Code Generation Agent output — Retail Store Task Execution #1'. Below the title, it says 'csdmichael wants to merge 37 commits into main from feature/agent/e724c65d'. A summary bar shows 'Conversation 0', 'Commits 37', 'Checks 5', and 'Files changed 37' with a net change of '+1,278 -1'. A comment from 'csdmichael' is visible, stating: 'Generated by the Agentic SDLC code generation agent for project Retail Store Task Execution (artifact e724c65d-59e5-46d8-be39-d78a1933a1a9). Scaffolds ui/ , api/ , db/ , and docs/ plus CI and Azure deployment workflows (37 files).' To the right of the comment, there are sections for 'Reviewers' (No reviews), 'Assignees' (No one assigned), and 'Labels' (None yet).

15. Accept test evidence

Test Planning created ADO Test Plan 426. Testing and Test Automation referenced successful GitHub Actions CI run 32870761400 for the exact feature-branch SHA. The same ADO plan was reused during the approval retry rather than duplicated.

Human review workspace

Edit proposals and select exactly what can be published. Saving never writes to ADO or GitHub.

3 awaiting review

☒ Include in approval

• Test Planning Agent

• Awaiting Approval

Approve & publish

Proposal title

Test Planning Agent output

Editable proposal

Retail Store Task Execution – Test Plan (Reviewable Draft)

This test plan is risk-based and traces to the approved requirements for the Retail Store Task Execution project. It

Summary view

Show full proposal

Expand all

Collapse all

Select all

Clear all

+ Add work item

Remove selected

> ☒ Section Retail Store Task Execution – Test Plan (Reviewable Draft)

> ☒ Section 1. Traceability Matrix

> ☒ Section 2. Test Areas, Risks, and Approach

3 of 3 items selected

Save draft

16. Approve security and compliance

The Security and Compliance Agent provides an executive release assessment with prioritized findings, control evidence, mitigations, residual risk, and a release recommendation. Security output remains proposal-only until this gate approves it.

Agentic SDLC — Software Factory

Agentic SDLC

Software Factory

AGENTIC SDLC OPERATIONS

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User Management

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Agent Configuration

Human review workspace

Edit proposals and select exactly what can be published. Saving never writes to ADO or GitHub.

☒ Include in approval

Security and Compliance Agent

Proposal title

Security and Compliance Agent output

Editable proposal

**Retail Store Task Execution – Security & Compliance Release Assessment (Dev I

Expand all

Collapse all

Select all

Clear all

+ Add work item

> ☒ Section 1. **Summary & Recommendation**

> ☒ Section 2. **Prioritized Findings**

> ☒ Section 3. **Evidence & Control Review**

17. Approve release and deployment

Before approval, PR 1 was open, clean, mergeable, and backed by successful CI. The DevOps / Release Agent proposed Azure resources, OIDC configuration, smoke checks, rollback, and go/no-go criteria.

After release, Ops Monitoring defines service indicators, alerts, dashboard views, incident triage, ownership, rollback signals, and a first-week checklist. Knowledge Assistant produces the support brief. Insights defines measurable adoption and reliability thresholds plus an ordered 30-day improvement backlog.

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Edit proposals and select exactly what can be published. Saving never writes to ADO or GitHub.

Include in approval

Ops Monitoring Agent

Proposal title

Ops Monitoring Agent output

Editable proposal

**Post-Release Operations Assessment: Retail Store Task Execution (Dev Environr

_Based on verified signals: Hosted UI, API health endpoint, Swagger UI, and Ope

Deployment run 32873630512 succeeded. All analysis is grounded in the approv

Expand all

Collapse all

Select all

Clear all

+ Add work item

>

Section

1. Service Health Indicators

>

Section

2. Actionable Alerts

>

Section

3. Dashboard Views

19. Confirm the completed project

All lifecycle stages and all nine approval decisions are complete. The project record remains the starting point for evidence and operations.

AGENTIC SDLC — Software Factory

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Retail Store Task Execution

Prioritised store task execution with photographic proof, offline capture, exception handling, and s

Approved

Overview

Intake details captured when the project was created.

BUSINESS OWNER	IT OWNER	ENVIRONMENT
myaacoub@microsoft.com	myaacoub@microsoft.com	Dev
AZURE DEVOPS	ADO VISIBILITY	GITHUB REPO
Retail Store Task Execution	private	retail-store-task-exec

12-agent SDLC workflow

Named Microsoft Foundry agents generate proposals through APIM. Human checkpoints publis

1. Requirements Agent

Agent action · plan · requirements-agent

Published

Last run 8/25/26, 4:31 AM

Back

Score

2. Planning Agent

Agent action · plan · planning-agent

Published

Back

The final five agents retain their Foundry identity, completion state, last run, checkpoint, and prerequisite context.

Agentic SDLC — Software Factory

Agentic SDLC

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8. Security and Compliance Agent

Agent action · security · security-compliance-agent

Published

Last run 8/25/26, 9:36 AM

9. DevOps / Release Agent

Agent action · deploy · devops-release-agent

Published

Last run 8/25/26, 9:41 AM

10. Ops Monitoring Agent

Agent action · operate · ops-monitoring-agent

Published

Last run 8/25/26, 9:55 AM

11. Knowledge Assistant

Agent action · operate · knowledge-assistant

Published

Last run 8/25/26, 9:55 AM

12. Insights Agent

Agent action · improve · insights-agent

Published

Last run 8/25/26, 9:56 AM

Provisioned systems of record

Created when the project was submitted. Retry after adding a missing credential.

Azure DevOps project

created

private

20. Verify all nine gates

The durable gate table records approver, role, selected artifacts, decision, and status. A closed gate cannot silently rerun its publication action. App Owners can reopen a gate only with an audit comment.

Agentic SDLC
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Approval gates

9 gate(s) · decisions are recorded on the audit trail.

Decision comment (optional)

Explain the decision for the audit record

Delegate review to

person@compan

GATE	STAGE	REQUIRES	LAST DECISION	ARTIFACTS
Plan and Scope Approval	Plan	business_user	approve by myaacoub@microsoft.com	Not required
Architecture and Design Approval	Design	it_user	approve by myaacoub@microsoft.com	• Architecture
Backlog Generation Approval	Plan	business_user	approve by myaacoub@microsoft.com	• Requirement • Planning Ag
Code Generation Approval	Build	it_user	approve by myaacoub@microsoft.com	• Code Gener
Pull				

Published result

Retail operator application

The deployed Angular 18 and Ionic 8 UI loads the generated Retail execution records from the published FastAPI service.

Retail Store Task Execution				
1	2	3	4	TASK IS
Sample Execution 1 E-0001				
Sample Execution 2 E-0002				
Sample Execution 3 E-0003				
Sample Execution 4 E-0004				

FastAPI and OpenAPI contract

Swagger exposes `/health` plus list, create, get, update, and delete operations for `/api/executions` .

Retail Store Task Execution API 0.1.0 OAS 3.1

/openapi.json

REST API for Retail Store Task Execution. Generated by the Agentic SDLC workflow.

health Liveness probe used by the deploy pipeline.

GET /health Health

executions Executions managed by this service.

GET /api/executions List Items

POST /api/executions Create Item

GET /api/executions/{item_id} Get Item

PATCH /api/executions/{item_id} Update Item

Azure DevOps project wiki

The DevOps / Release Agent now ensures one project wiki and upserts `/Project-Overview` after publication. The page includes:

- project name and description;
- hosted UI, API, health, Swagger, and OpenAPI links;
- GitHub repository, released pull request, deployment workflow/run, and merge commit;
- Business and IT ownership;
- selected UI, API, and data stacks; and
- the human-governance statement.

Release retries update the same page with its current ETag instead of creating duplicates. Resources discovered as pre-existing are never treated as workflow-owned deletion targets.

Operate and audit

Agent Activity

Agent Activity correlates every run with its lifecycle stage, status, model route, artifacts, and correlation identifier.

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Agentic SDLC

Software Factory

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Agent Activity

Every agent run with its model, APIM route, token estimate, guardrail findings, and correlation ID.

AGENT	MODEL & ROUTE
insights-agent ## Post-Release Insights & Continuous Improvement Proposal: Retail Store Task Execution ### 1. Measurable Metrics Definition **Adoption** - **Store Activation Rate:**% of stores with at least one task completed in the app per week. - **Colleague Engagement:** Average number of unique colleagues completing tasks per store per week. **Task Completion** - **On-Time Completion Rate:** % of tasks c	gpt-4.1 foundry- agents{projectPath}/agents/{
knowledge-assistant **Operational Knowledge Brief: Retail Store Task Execution (Dev Environment)** --- **Purpose** Enable prioritized, traceable store task execution with photographic proof, offline capture, exception handling, and same-day district compliance visibility. Designed to replace email/PDF-based tasking with a live, actionable, and auditable workflow for stores and district managers. _Evidence: [Req	gpt-4.1 foundry- agents{projectPath}/agents/{
ops-monitoring-agent **Post-Release Operations Assessment: Retail Store Task Execution (Dev Environment)** _Based on verified signals:	

Audit Trail

Authentication, agent execution, artifact edits, decisions, delegations, connector writes, merges, publication, and gate recovery are recorded with actor and correlation context.

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Audit Trail

Append-only record of user actions, agent actions, and approval decisions.

Filter by action, actor, or target

ACTOR	ACTION	TARGET
user myaacoub@microsoft.com	approval.approve	approval ffe472a
user myaacoub@microsoft.com	artifact.publish.approved	artifact 6245868
user myaacoub@microsoft.com	artifact.publish.approved	artifact d54b77b
user myaacoub@microsoft.com	auth.otp.verify	auth myaacoul
agent insights-agent	agent.run.complete	agentRun f144824

Approval evidence rules

Gate	Minimum evidence	What approval unlocks or publishes
Plan and Scope	Submitted project and intake documents	Requirements and Planning runs
Backlog Generation	Selected Planning Agent artifact	ADO hierarchy, sprints, queries, dashboards, delivery plan
Architecture and Design	Selected Architecture artifact	Versioned docs/architecture-design.md and Code Generation
Code Generation	Selected Code Generation artifact	Generated branch and pull request
Pull Request Review	GitHub repository evidence with pullRequestUrl	Security/Release prerequisite
Test Acceptance	Test Planning, Testing, and Test Automation artifacts plus successful run evidence	Security review
Security Review	Selected Security and Compliance artifact	Release prerequisite
Release and Deployment	Selected DevOps / Release artifact, reviewed PR, test, and security evidence	Merge, Azure publish, ADO wiki, post-release agents

Gate	Minimum evidence	What approval unlocks or publishes
Operate and Improve	Selected Ops Monitoring and Insights artifacts	Completed operating lifecycle

Release identity prerequisite

Generated repositories use GitHub OIDC and do not store an Azure client secret. The configured Entra application must trust the exact ID-qualified repository/environment subject:

```
repo:<owner>@<ownerId>/<repository>@<repositoryId>:environment:production
```

The API managed identity needs ownership of the OIDC application and the approved Microsoft Graph permission to maintain only credentials for applications it owns. If an authorized Entra owner pre-creates a verified repository credential for recovery, `GITHUB_FIC_PREPROVISIONED=1` may be enabled only for that release and must be removed immediately afterward.

Expected traceability chain

A completed project leaves:

- three versioned intake documents in GitHub;
- approved requirements and architecture documents;
- a reviewed ADO backlog and planning assets;
- generated application code and tests behind a pull request;
- accepted CI, test-plan, and security evidence;
- one approved merge and one deployment workflow;
- live UI, API health, Swagger, and OpenAPI endpoints;
- an idempotent ADO Project Overview wiki page;
- post-release operations and improvement proposals; and
- nine attributable human decisions in the audit trail.

Rebuild this guide

The Markdown file is the source of truth. After changing guide content or UI captures, rebuild and validate the PDF with installed Chrome:

```
api/.venv/Scripts/python.exe scripts/build_user_guide.py
```

The builder resolves local images, creates a print-safe table of contents, exports `Agentic-SDLC-User-Guide.pdf`, and verifies the result with `pypdf`.